

Skill-10

```
from shell.c:5:
/usr/include/linux/limits.h:11: note: this is the location of the previous definition
11 | #define MAX_INPUT 255 /* size of the type-ahead buffer */
    |
=====
ENVIRONMENT MINI SHELL
=====
myshell$ pwd
/home/himasree/OSLab
myshell$ export NAME=Himasree
Exported NAME=Himasree
myshell$ echo $NAME
Himasree
myshell$ export CITY=Hyderabad
Exported CITY=Hyderabad
myshell$ export CITY=Bangalore
Exported CITY=Bangalore
myshell$ echo $CITY
Bangalore
myshell$ export 123CITY=India
export: invalid variable name: 123CITY
myshell$ export NAME
export: invalid syntax
Usage: export NAME=value
myshell$
```

```
Exiting shell...
himasree@LAPTOP-41COT5AA:~/OSLab$ cat shell.c
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>
#include <limits.h>
#include <ctype.h>

#define MAX_INPUT 1024
#define MAX_ARGS 100

/* =====
PWD - CURRENT DIRECTORY
===== */

int builtin_pwd(char **args)
{
    char cwd[PATH_MAX];

    /* Retrieve current directory */
    if (getcwd(cwd, sizeof(cwd)) == NULL) {
        perror("pwd");
        return 1;
    }

    /* Display path */
    printf("%s\n", cwd);
}
```

```

/* =====
VALIDATE ENVIRONMENT VARIABLE NAME
===== */

int valid_name(const char *name)
{
    if (name == NULL || name[0] == '\0')
        return 0;

    /* First character must be letter or underscore */
    if (!isalpha((unsigned char)name[0]) &&
        name[0] != '_')
        return 0;

    /* Remaining characters */
    for (int i = 1; name[i] != '\0'; i++) {
        if (!isalnum((unsigned char)name[i]) &&
            name[i] != '_')
            return 0;
    }

    return 1;
}

/* =====
EXPORT
===== */

```

```

int builtin_export(char **args)
{
    /* No argument */
    if (args[1] == NULL) {
        printf("Usage: export NAME=value\n");
        return 1;
    }

    char *assignment = args[1];

    /* Find '=' */
    char *equal = strchr(assignment, '=');

    if (equal == NULL) {
        printf("export: invalid syntax\n");
        printf("Usage: export NAME=value\n");
        return 1;
    }

    /* Separate NAME and VALUE */
    *equal = '\0';

    char *name = assignment;
    char *value = equal + 1;

    /* Validate variable name */

```

```
    if (strlen(input) == 0)
        continue;

    /* Parse command */
    tokenize(input, args);

    if (args[0] == NULL)
        continue;

    /* Find built-in */
    builtin_function command =
        find_builtin(args[0]);

    if (command != NULL) {

        /*
         * Execute in current process.
         */
        command(args);

    } else {

        printf("myshell: command not found: %s\n",
            args[0]);

    }

}

return 0;
}
```

himasree@LAPTOP-41COT5AA:~/OSLab\$ |